Experiment No. 1

STUDY OF LOGIC GATES

AIM: To familiarize

i. TTL and CMOS Logic ICs for AND, OR, NOT, NAND, NOR and XOR by verification of truth tables.

COMPONENTS AND EQUIPMENTS REQUIRED:

SL No.	COMPONENTS	SPECIFICATION	QTY
1.	AND GATE	IC 7408	1
2.	OR GATE	IC 7432	1
3.	NOT GATE	IC 7404	1
4.	NAND GATE 2 I/P	IC 7400	1
5.	NOR GATE	IC 7402	1
6.	X-OR GATE	IC 7486	1
7.	IC TRAINER KIT	-	1
8.	Connecting Wires	As required	

PRINCIPLE:

Circuit that takes the logical decision and the process are called logic gates. Each gate has one or more input and only one output. OR, AND and NOT are basic gates

AND GATE:

The AND gate performs a logical multiplication commonly known as AND function. The output is high when both the inputs are high. The output is low level when any one of the inputs is low. Output Y= A.B where A and B are the inputs.

OR GATE:

The OR gate performs a logical addition commonly known as OR function. The output is high when any one of the inputs is high. The output is low level when both the inputs are low. Output Y= A+B where A and B are the inputs.

NOT GATE:

The NOT gate is called an inverter. The output is high when the input is low. The output is low when the input is high. Output Y= NOT(A) where A is the input.

NAND GATE:

The NAND gate is a combination of AND-NOT. The output is high when any one of the input is low or both inputs are low .The output is low level only when both inputs are high.

NOR GATE:

The NOR gate is a combination of OR-NOT. The output is high when both inputs are low. The output is low when one or both inputs are high.

X-OR GATE:

The output is high when any one of the inputs is high. The output is low when both the inputs are low and both the inputs are high.

PROCEDURE:

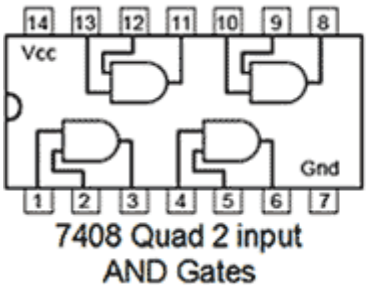
1. Check the all the required components.
2. Connect the circuit as per the circuit diagram.
3. Apply Vcc and Ground.
4. Logical inputs are given as per truth table using switches provided.
5. Observe the output on LEDs and verify the truth table.

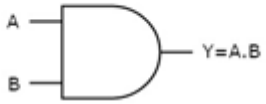
RESULT:

Familiarized with basic TTL and CMOS logic gate ICs and verified their truth tables.

AND GATE

PIN DIAGRAM **SYMBOL** **TRUTH TABLE**

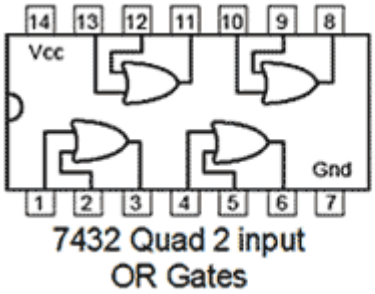


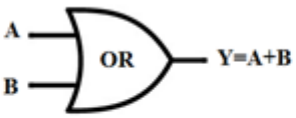


A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

OR GATE:

PIN DIAGRAM **SYMBOL** **TRUTH TABLE**

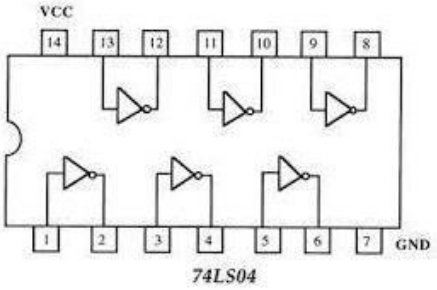


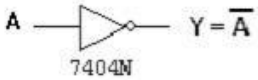


A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

NOT GATE:

PIN DIAGRAM **SYMBOL** **TRUTH TABLE**

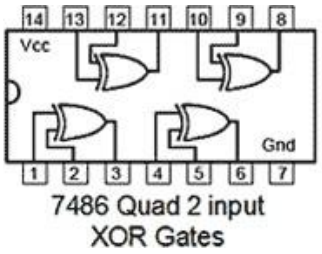




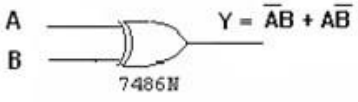
A	$\overline{A}$
0	1
1	0

X-OR GATE :

PIN DIAGRAM **SYMBOL** **TRUTH TABLE**



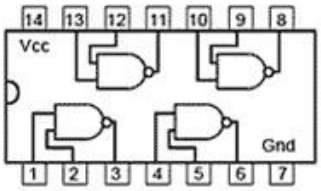
7486 Quad 2 input
XOR Gates



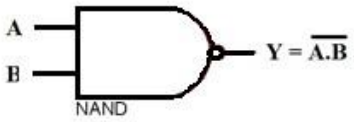
A	B	$\overline{A}B + A\overline{B}$
0	0	0
0	1	1
1	0	1
1	1	0

2-INPUT NAND GATE:

PIN DIAGRAM **SYMBOL** **TRUTH TABLE**



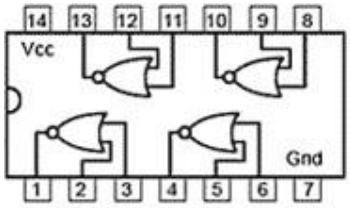
7400 Quad 2 input
NAND Gates



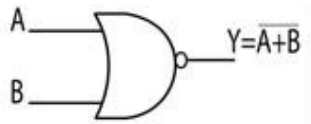
A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

NOR GATE:

PIN DIAGRAM **SYMBOL** **TRUTH TABLE**



7402 Quad 2 input
NOR Gates



A	B	Y= $\overline{A+B}$
0	0	1
0	1	1
1	0	1
1	1	0